Mgcingane Lihle Ahlume

SQL Fundamentals: SELECT & FILTERING

a) Selecting & Sorting

Table employees

ID	first_name	last_name	department	salary	hire_date	city
1	Alice	Green	IT	70 000	2020-01-10	Johannesburg
2	Brian	Lee	HR	45 000	2019-03-22	Cape Town
3	Cathy	Zulu	Finance	65 000	2018-07-18	Durban
4	David	Mokoena	Marketing	50 000	2021-11-05	Pretoria
5	Eva	Naidoo	IT	72 000	2017-09-30	Johannesburg

1. SELECT *
   FROM employees;

ID	first_name	last_name	department	salary	hire_date	city
1	Alice	Green	IT	70 000	2020-01-10	Johannesburg
2	Brian	Lee	HR	45 000	2019-03-22	Cape Town
3	Cathy	Zulu	Finance	65 000	2018-07-18	Durban
4	David	Mokoena	Marketing	50 000	2021-11-05	Pretoria
5	Eva	Naidoo	IT	72 000	2017-09-30	Johannesburg

2. SELECT DISTINCT department
   FROM employees;

department
IT
HR
Finance
Marketing

3. SELECT first_name,
last_name
FROM employees
ORDER BY salary DESC;

first_name	last_name
Eva	Maidoo ~~72 000~~
Alice	Green
Cathy	Zulu
David	Mokoena
Brian	Lee

4. SELECT ID,
first_name,
last_name,
salary
FROM employees
ORDER BY salary DESC
LIMIT 3;

ID	first_name	last_name	salary
5	Eva	Maidoo	72 000
1	Alice	Green	70 000
3	Cathy	Zulu	65 000

02) Filtering with WHERE

5).
```
SELECT ~~to~~ ID,
       first_name,
       last_name,
       department
FROM employees
WHERE department = 'IT';
```

ID	First-name	last-name	department
1	Alice	Green	IT
5	Eva	Maidoo	IT

6.
```
SELECT ID,
       first_name,
       last_name,
       department,
       salary
FROM employees
WHERE department = 'Finance' AND salary > 60 000;
```

ID	first-name	last-name	department	salary
3	Cathy	Zulu	Finance	65 000

7.
```
SELECT ID,
       first_name,
       last_name,
       department
FROM employees
~~WHERE department = 'HR' OR 'Marketing';~~
WHERE department = 'HR' OR department = 'Marketing';
```

ID	first_name	last_name	department
2	Brian	Lee	HR
4	David	Mokoena	Marketing

3.
```
SELECT  ID,
        first_name,
        last_name,
        department
FROM    employees
WHERE   department NOT IN ('IT');
```

ID	first_name	last_name	department
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
4	David	Mokoena	Marketing

4.
```
SELECT  ID,
        first_name,
        last_name,
        department
FROM    employees
WHERE   department IN ('IT','HR','Finance');
```

ID	first_name	last_name	department
1	Alice	Green	IT
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
5	Eva	Maidoo	IT

10. SELECT ID,
first-name,
last-name,
department,
salary,
city
FROM employees
WHERE department IN ('IT') AND salary > 65 000 AND city = 'Johannesburg';

ID	first-name	last-name	department	salary	city
	Alice	Green	IT	70 000	Johannesburg